

BudgetSavvy

Project Overview

BudgetSavvy is a SQL-based budget recommendation system developed to help users identify relevant Union Budget 2025 initiatives based on demographic attributes such as profession, age group, and location. The project focuses on efficient storage, organization, retrieval, and filtering of policy information through a structured relational database.

The code for the project can be found here :

<https://github.com/aishwaryaSingh2804/BudgetSavvy>

Technical Skills Demonstrated

1. Database Design and Data Modeling

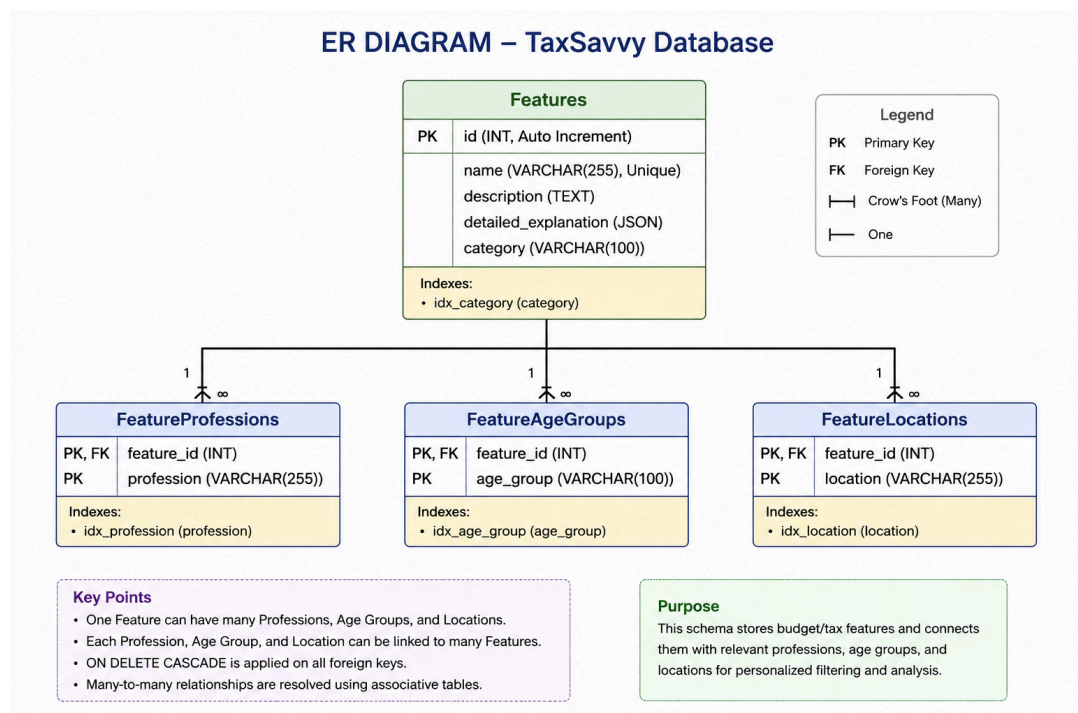
Designed a normalized relational database schema to organize budget features and policy information efficiently.

Skills Demonstrated:

- Entity Relationship Modeling
- Database Normalization
- Primary Key and Foreign Key Design
- Many-to-Many Relationship Modeling
- Schema Design

Tables Created:

- Features
- FeatureProfessions
- FeatureAgeGroups
- FeatureLocations



ER Diagram BudgetSavvy

2. Relational Database Management

Implemented relationships between budget features and demographic categories to support personalized recommendations.

Skills Demonstrated:

- Relational Database Design
- Data Integrity Management
- Referential Integrity Constraints
- Junction Table Implementation

Business Value:

Allows a single budget feature to be associated with multiple professions, age groups, and locations.

3. SQL Query Development

Developed SQL queries to retrieve relevant budget features based on multiple filtering criteria. Skills Demonstrated:

- Complex SQL Query Writing
- Dynamic Filtering
- Subqueries
- Conditional Query Construction
- Data Retrieval Optimization

Business Value:

Users can quickly identify policies relevant to their demographic profile.

4. Stored Procedures and Automation

Developed the AddFeatureWithTags stored procedure to automate insertion of new budget features and associated demographic tags.

Skills Demonstrated:

- Stored Procedure Development
- Database Automation
- String Parsing in SQL
- Data Validation
- Process Optimization

Business Value:

Reduces manual effort and minimizes insertion errors during data onboarding.

5. Query Optimization

Implemented indexing on frequently filtered attributes.

Skills Demonstrated:

- Query Performance Optimization

- Indexing Strategies
- Database Tuning

Indexes Created:

- Category Index
- Profession Index
- Age Group Index
- Location Index

Business Value:

Improves data retrieval speed for filtering operations.

6. Semi-Structured Data Management

Utilized JSON columns to store detailed policy explanations and impact descriptions.

Skills Demonstrated:

- Structured and Semi-Structured Data Handling
- JSON Data Types in SQL
- Flexible Data Storage Design

Business Value:

Supports storage of rich policy descriptions without excessive table complexity.

7. Data Segmentation and Recommendation Logic

Designed the database structure to support demographic

segmentation. Skills Demonstrated:

- User Segmentation
- Data Categorization
- Recommendation Logic Design
- Business-Oriented Data Modeling

Business Value:

Enables personalized policy discovery for different user groups.

Key Learning Outcome

This project strengthened my understanding of how data is structured, organized, optimized, and retrieved in real-world systems. It provided hands-on experience in designing scalable databases, automating data workflows, and developing efficient querying solutions that support data-driven decision-making.